



CASE STUDY | BEGINNER DATA ANALYSIS

How Is the Shop Performing?

The Scenario

You have just joined an online shop that sells electronics, accessories, wearables, home office items, stationery and gaming products. The **Head of Operations** wants a clear picture of how the shop has been performing from January 2024 to June 2026. They are not a data person, so they need simple numbers, clear charts and plain-language explanations. Your job is to turn four raw data files into that picture.

Your Deliverable

A short presentation or dashboard (5 to 8 slides or one dashboard page) that answers the business questions in Step 5 and ends with **three recommendations** the Head of Operations can act on. You may use Excel, Power BI, Google Sheets, SQL or Python.

The Data

File	What it contains	Key columns
customers.csv	One row per customer (10,000 rows)	CustomerID, Age, City, SignupDate, CustomerSegment
orders.csv	One row per order line (about 50,000 rows)	OrderID, CustomerID, OrderDate, ProductID, Quantity, Discount, PaymentMethod, Status
payments.csv	One row per payment attempt	PaymentID, OrderID, PaymentDate, PaymentStatus
products.csv	The 20 products the shop sells	ProductID, ProductName, Category, UnitPrice

How the files connect: orders link to customers through **CustomerID**, to products through **ProductID**, and to payments through **OrderID**. Draw this on paper before you start.

Steps to Follow

Step 1 Get to know the data

Open each file and look before you calculate anything.

- How many rows and columns does each file have? What does one row represent?
- Which columns are numbers, which are text and which are dates?
- Write down one question each file could help you answer.

Step 2 Clean the data

Real data is messy, and this data is no exception. Check for each of the problems below and write down what you found and what you decided to do about it.

- Missing values: which columns have blanks? Should you remove, fill or keep those rows?
- Duplicates: are any orders recorded more than once?
- Values that make no sense: look at the smallest and largest numbers in each numeric column. Do the product prices look sensible for each item?
- Inconsistent text: are the same cities spelled in more than one way?
- Broken links: does every order point to a customer and product that actually exist?

Step 3 Combine the tables

Join the orders table to products (to get the price and category), to customers (to get city and segment) and to payments (to get payment status). Check that your row count still makes sense after each join.

Step 4 Create new columns

Some useful numbers are not in the data yet, so you must build them.

- **Revenue** for each order: $\text{Quantity} \times \text{UnitPrice} \times (1 - \text{Discount})$.
- **Year** and **Month** from OrderDate, so you can look at trends over time.
- Think carefully: should cancelled, returned or unpaid orders count as revenue? Decide, and explain your choice.

Step 5 Answer the business questions

Use totals, averages, counts and percentages. Group your data (a pivot table works well) to answer:

- How much revenue did the shop make, from how many orders? What is the average order value?
- Is revenue growing, shrinking or flat month by month? Are there any seasonal peaks?
- Which products and categories bring in the most revenue? Which sell the most units? Are these the same?
- Which cities and customer segments (New, Regular, VIP) are most valuable?
- What share of orders are cancelled or returned? What share of payments fail? Do some payment methods fail more often?
- Do bigger discounts lead to bigger orders, or just lower revenue?

Step 6 Visualise your findings

Choose the right chart for each question. Keep every chart clean: one message, a clear title, labelled axes, no clutter.

- **Line chart** for trends over time. **Bar chart** for comparing products, categories or cities.
- **KPI cards** (big single numbers) for headline figures such as total revenue and number of orders.
- Avoid pie charts with more than four or five slices.

Step 7 Tell the story

Write for the Head of Operations, not for another analyst.

- Start with a one-slide summary of the three most important findings.
- For each chart, write one sentence that says what it means ("So what?"), not just what it shows.
- Finish with three practical recommendations backed by your numbers, plus a short note on the data problems you found and how you handled them.

Submission Checklist

- Your cleaned dataset or working file, and a short list of the cleaning decisions you made.
- Your presentation or dashboard with the summary, charts and three recommendations.
- Every number in your presentation can be traced back to your working file.

Tip: if you get stuck, go back to the question you are trying to answer. Every calculation should serve a question.